

History-Dependent Ontology (HDO)

A Philosophical Interpretation of History-Dependent Dynamics (HDD)

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Preface

HDD asks a scientific question: When does a system's history make a difference to its future?

HDO gives the philosophical answer: When that history reorganizes the system's available transitions, it becomes part of what the system is.

The Bridge Principle governs the relation: HDD establishes empirical facts. HDO interprets their ontological significance. The inference from “history is dynamically relevant” to “history is constitutive of processual identity” is abductive — it is a highly coherent and well-motivated interpretation of the evidence, not a deductive consequence.

HDD's methodological neutrality toward ontology is a deliberate and permanent feature of that framework, not a placeholder awaiting completion by HDO. HDO does not revise HDD's stance within HDD's own domain, and accepting or rejecting HDO's interpretation has no bearing on HDD's standing as a methodology: a reader can adopt every empirical construct and evidential procedure in HDD while declining everything argued for here. HDO opens a separate, explicitly labeled interpretive layer alongside HDD — it does not walk back HDD's bracketing of ontology, because that bracketing was never a promise to eventually take a side.

This epistemic modesty is structural to HDO, not a rhetorical hesitation.

1. What HDD Establishes (Summary)

HDD distinguishes five scientific constructs:

History-Dependent Predictive Structure (C-I) — the past improves prediction.

Causal Trajectory Dependence (C-II) — manipulating the past alters the future.

Feedback (C-III) — causal cycles from system effects to future dynamics.

Functional Self-Reference (C-IV) — the system uses a representation of itself.

Self-Modeling (C-V) — the system predicts its own future states, with error influencing behavior.

Central Principle: Evidence for one construct does not authorize inference to the next.

Level	Question	Status
Predictive	Does the past improve prediction?	Empirical
Causal	Does manipulating the past alter the future?	Empirical
Organizational	Has history altered the organization of available transitions?	Empirically investigable

Level	Question	Status
Constitutive	Is this historicity constitutive of what the process is?	Philosophically interpreted

Table 1. Levels of inference from HDD findings to HDO interpretation.

Validation status (added for this edition): as of the current HDD deposit (Zenodo, DOI 10.5281/zenodo.21955745, published August 2026), only Construct I (predictive history dependence) has been empirically validated, via a computational benchmark across four synthetic systems. Constructs II–V are proposed extensions that the deposit itself describes as requiring independent experimental validation. Table 1's "Empirical" and "Empirically investigable" labels should be read accordingly: they mark HDD's methodological commitments, not results already in hand for every construct. Readers should not infer that C-II–C-V are established findings on the strength of C-I's validation.

For full methodological details, refer to the original HDD publications.

2. The Core Thesis

HDO's central claim is this:

When historical organization changes the space of dynamically available transitions through which a process can continue, history ceases to be merely causal background. It becomes constitutive of what the process is.

This is not offered as a metaphor. It is an ontological claim about the nature of processes — held with conviction, and argued for below, but interpretive in status per the Bridge Principle above.

The process is not exhausted by its instantaneous state. Its identity is not reducible to its present configuration. What it is includes what it has become — and what it has become is written not as memory stored somewhere, but as the very structure of its ongoing organization.

Signature Proposition: When historical organization changes which transitions remain dynamically available, history is no longer merely something that happened to the process; it is part of the organization through which the process continues to exist and transform.

On the Abductive Character of the Claim

This is not a claim that follows deductively from HDD's findings. It is an interpretation — but an interpretation that is constrained, motivated, and rendered plausible by those findings. The move from “history reorganizes available transitions” to “history is constitutive of processual identity” is abductive: it is the reading that best unifies the phenomena under a single ontological vocabulary.

Alternative readings remain viable: history as accumulated constraint without constitutive status; history as path-dependent boundary condition that does not enter identity; history as causal background that

happens to be durable. HDO does not claim these alternatives are impossible. It claims that when history reorganizes the space of available transitions, treating that history as constitutive of identity provides a more coherent account than treating it as external.

3. Why “History-Dependent”?

This is not generic process philosophy. This is a specific ontological claim about the role of history.

Generic process philosophy says: everything is process. From Heraclitus to Whitehead, the insight that “what most fundamentally exists is change, or process” has been a recurring theme in Western thought.

HDO says: processes are constituted by their history.

The difference is decisive. A process could be purely recurrent, returning to the same states in the same cycles — a pendulum, a tide, a clock. That is process, but it is not history-dependent process. Its past does not reorganize its future.

HDO targets processes whose history does not merely precede them but persists in them — as reorganized transition spaces, altered possibilities, transformed organization. This is the distinctive claim: not that everything is process, but that some processes are historically constituted.

4. Why This Matters

The default ontology of the modern West is substance ontology: reality is made of things that persist through time, retaining identity across change. This metaphysics, rooted in Aristotle and reinforced by classical physics, has been described as “inappropriate” for understanding the living world.

What persists is not substance — it is organization. A flame persists. A cell persists. A mind persists. All of them are processes, not things. They maintain their form through continuous transformation. Their identity is not in their material composition but in the historically organized dynamics through which they continue to exist.

As philosopher of biology John Dupré argues, what we are tempted to think of as constant things are “in reality merely temporary stabilities in this constant flux of change, eddies in the flux of process” (Dupré, 2020). A process ontology, he contends, is “the only kind adequate to making sense of the living world” (Dupré, 2020).

Metaphysics	View of History	View of Identity
Naïve substance	History is external to identity	Identity is static persistence of substance
Sophisticated substance	History may shape identity, but is not constitutive	Identity is continuity of form or organization

Metaphysics	View of History	View of Identity
HDO	History is constitutive of identity	Identity is continuity of historically organized transition structure

Table 2. Comparison of metaphysical positions on history and identity.

HDO is strongest against the naïve view. Against sophisticated substance views, the disagreement is more subtle: HDO claims that history is not merely shaping identity but part of what identity is. Borderline cases may blur this distinction — for example, a sophisticated substance view that treats “form” as historically generated may, in practice, converge with HDO's account. The difference is one of emphasis: whether history is treated as a constitutive component of identity or as a generative process that produces identity but does not remain part of it.

5. The Ontological Position

5.1 Contemporary Convergence

Recent research provides several empirically distinct lines of work that are compatible with this view — not as proof, but as converging points of interpretive relevance.

Ilya Prigogine's theory of dissipative structures provides a concrete physical case of the kind of history-dependent organization HDO is concerned with. Prigogine showed that systems far from thermodynamic equilibrium can spontaneously form ordered structures — not despite their openness to the environment, but because of it. These structures are defined by their history: the specific organization that emerges depends on the trajectory the system took to its current conditions (Prigogine & Stengers, 1984; Prigogine, 1980).

More importantly, Prigogine's concept of bifurcation describes a class of physical situations in which history has exactly this kind of effect. At a bifurcation point, the system's future is not determined by its present state alone: small fluctuations decide which branch the system takes, and once taken, the space of available transitions is reorganized — some possibilities close, others open. This is a concrete physical case of the kind of history-dependent reorganization that HDO interprets as constitutive — not as a proof of the ontological claim, but as a physical situation in which prior trajectories can alter the system's subsequent dynamical possibilities.

Prigogine also argued that irreversibility is fundamental to nature, not a subjective illusion. Prigogine's account of irreversibility strengthens the physical case for treating temporal history as dynamically consequential. It does not, by itself, establish the stronger HDO claim that history is constitutive of processual identity — that further step is the interpretive move HDO makes, not a conclusion Prigogine's physics forces on its own (Prigogine, 1980).

Michael Levin demonstrates that goal-directed competency can occur without centralized nervous systems. Agency is a property of organizations, not substances. His research program on basal cognition adopts an evolutionary perspective for studying biological cognition across diverse embodiments, showing

that mind is a “scalable, substrate-independent property of living matter” (Levin, 2026). This does not prove process ontology, but it renders it intelligible: it shows that the phenomena associated with agency and cognition can be studied across embodiments without presupposing a privileged substrate.

Important caveat on Levin: The citation above draws on Levin's empirical work on basal cognition — goal-directed competency without centralized nervous systems — which is directly relevant to HDO's claims about organization and agency. However, Levin's “Ingressing Minds” hypothesis proposes that cognitive patterns exist in a structured, non-physical “Platonic space” from which they “ingress” into physical embodiments. This is, on a natural reading, closer to a contemporary Platonism than to a purely immanent process ontology — it posits patterns that are not themselves physical processes, which sits in tension with HDO's claim that organization is always the organization of a physical, historically constituted process. The citation here endorses Levin's empirical findings independently of that further metaphysical hypothesis, and should not be read as an endorsement of the Platonic-space framework.

Karl Friston's Free Energy Principle formalizes self-organization as the maintenance of dynamical states against dispersion. He argues that “life — or biological self-organization — is an inevitable and emergent property of any (ergodic) random dynamical system that possesses a Markov blanket” (Friston, 2013). The boundary between system and environment is a statistical relation, not a substance. Friston's formalism provides a mathematical language for describing what HDO interprets philosophically: systems are defined not by what they are made of, but by the organization through which they maintain themselves.

Caveat on Friston: as with Levin, this citation is to Friston's formal framework and its interpretive value, not an endorsement that the framework is uncontested. A published technical critique (Biehl, Pollock & Kanai, 2020) identifies specific formal errors in the 2013 derivation, including in the argument connecting the Markov-blanket construction to active Bayesian inference. HDO's use of Friston here is at the level of vocabulary and orientation — systems defined by statistical boundary-maintenance rather than substance — not a claim that the 2013 proof is formally airtight.

Anil Seth's predictive processing shows that the experienced self arises from predictive processes. Seth's “controlled hallucination” perspective provides a framework for understanding how the brain constantly generates and updates a model to predict sensory input. The self is not a thing the brain contains; it is a pattern the brain maintains (Seth, 2021). Seth's work does not establish HDO, but it aligns with its direction: selfhood is a processual achievement, not a substantial given.

Eva Jablonka and Simona Ginsburg have argued for the evolutionary emergence of minimal consciousness through the accumulation of learning and memory capacities. Their work suggests that the capacity for open-ended learning and the flexible use of stored information may be a key step in the emergence of subjective experience (Ginsburg & Jablonka, 2019).

Giulio Tononi's Integrated Information Theory attempts to quantify consciousness in terms of integrated information. While HDO does not endorse IIT's specific metrics, the theory shares with HDO the view that consciousness is not a substance but a property of informational organization (Tononi, 2008).

Francisco Varela's work on autopoiesis describes living systems as networks of processes that produce their own components. The concept of autopoiesis provided a foundational critique of substance-based accounts of life, emphasizing instead the self-producing, self-maintaining organization of living systems (Maturana & Varela, 1980). Varela's later account of autopoietic identity describes it as held in a standing tension between the system's tendency to keep producing itself and its need to remain distinguishable from its environment — a tension between self-production and self-distinction, rather than a stable resolution of either (Varela, 1992).

These frameworks converge on a common picture: agency, organization, selfhood, and even consciousness are properties of organized dynamics, not of substances. None of them establishes HDO. They make it a coherent, well-anchored interpretation.

A caveat on “convergence”: the word should not be read as implying these frameworks agree with one another. IIT, the Free Energy Principle, and predictive processing are not variants of a single theory — they disagree on substantive points, including how integration relates to consciousness and whether a single quantitative measure of consciousness is coherent at all. What converges across them is narrower than it may appear: a shared rejection of substance-based accounts of selfhood and agency, not a shared positive theory. HDO draws on that narrower convergence, not on an underlying consensus that does not exist.

5.2 The Seven Theses

The following theses are stated in a more declarative register than Section 2's careful hedging. This is a deliberate shift, not an inconsistency: these are HDO's working commitments, held with conviction, not claims that have been independently established beyond the abductive case already made. Each thesis should be read with an implicit “HDO holds that...” even where the prose states it directly.

Thesis 1 — Processual Primacy

On HDO's reading, reality is processes, not substances. What we call “entities” are relatively stable patterns of process. A wave, a flame, a cell, a mind — all maintain their form through continuous transformation. Substance is treated as an abstraction from process, not the other way around.

Thesis 2 — Historicity

Processes carry their history as part of their organization. The past does not merely influence the present; on this view, it constitutes it — not as memory stored elsewhere, but as the structure of the process itself. The physical mechanism for this thesis is Prigogine's bifurcation: at a bifurcation point, the path taken reorganizes the space of available transitions, and that reorganization persists as part of the system's structure. Bergson made a closely related point about duration: no two moments of a conscious life are identical, because each moment carries forward something of what preceded it (Bergson, *The Creative Mind*). From that insight to the processual view that “the existence of things always depends on the existence of processes,” historicity is treated as a core feature of processual reality.

Thesis 3 — Relationality

A process is not self-contained. Its available states depend on its internal organization and external relations. On this view, it has no fixed “nature” independent of its relations. In processual ontology, “the interrelated elements exist nowhere else than in their interrelations” (Rescher, 2012).

Thesis 4 — Individuation

Individuals are treated as emerging from processes, rather than being presupposed by them. What we call an “individual” is, on this reading, a relatively stable resolution of a process of becoming. As Simondon argued, the process of individuation is biologically inseparable, but ontologically prior to the individual (Simondon, 1958/2005).

The Individuation Criterion: A process maintains identity across reorganization if the organization of its dynamically available transitions remains structurally continuous. A break in continuity may be indicated when historically organized causal dependence (C-II), feedback organization (C-III), functional self-reference (C-IV), or self-modeling (C-V) undergoes sufficiently large and persistent disruption. Identity under HDO is defined by the continuity of historically organized transition structure, and that continuity is empirically investigable through the HDD constructs — but the constructs remain analytically distinct evidential claims, and disruption of any one of them is treated here as indicative of a possible break in identity, not as individually necessary or sufficient for it.

Examples of continuity and discontinuity: An organism undergoing metamorphosis (caterpillar to butterfly) maintains a continuous causal trajectory (C-II), feedback loops (C-III), and self-referential organization (C-IV) despite dramatic morphological change — identity is preserved. A neural network that loses its self-modeling capacity (C-V) after a lesion, and whose feedback loops (C-III) are permanently broken, shows the kind of large, persistent disruption that HDO treats as indicating a break in the continuity of its historically organized transition structure — identity has changed. A collective system that undergoes an irreversible phase transition from ordered to disordered dynamics loses its causal trajectory dependence (C-II) and feedback organization (C-III) — identity has shifted. A stem cell lineage that differentiates into two distinct cell types loses the historical continuity of its available transitions — identity has bifurcated.

HDO inherits, in a processual register, the question Derek Parfit posed for personal identity: what actually matters in survival is psychological continuity and connectedness, not the metaphysical further fact of being “the same” substance (Parfit, 1984). HDO’s answer substitutes continuity of historically organized transition structure for Parfit’s continuity of psychological states, but the underlying move — replacing an all-or-nothing identity fact with a matter of degree — is the same.

A sharp threshold for identity is not missing; it is not the right kind of thing to look for. Asking exactly where C-III’s degradation crosses from “degraded” to “broken” presupposes a binary fact that substance ontology needs and HDO denies exists. The Individuation Criterion is reliably diagnostic at the extremes; in between, identity is expected to be a matter of degree. This is not a gap to be closed by future work. It is what a historically constituted identity should look like — a commitment HDO makes as an ontological interpretation, not a conclusion HDD’s empirical constructs establish on their own.

Thesis 5 — Self-Reference

On this view, in some processes the dynamics comes to represent and modify its own operation. Self-reference is treated as graded, not binary — and as a functional achievement, not a metaphysical given.

Thesis 6 — Openness

No finite process is treated as absolutely self-sufficient. Autonomy is not equated with isolation. Operational closure is not read as independence from the environment. Varela's later work on autopoiesis describes living systems as held open by exactly this tension — sustaining themselves as a distinct unity while remaining constitutively dependent on exchange with an environment (Varela, 1992).

Thesis 7 — Consciousness

Whether historical self-reference constitutes sufficient, necessary, or merely correlative conditions for phenomenal experience remains, on HDO's own terms, an open question. HDO does not claim to have solved consciousness — but it offers a framework for asking the question without presupposing the traps of substance ontology.

6. Lack: The Structural Non-Coincidence

If processes are historically constituted, then no process fully coincides with itself. This is not a psychological observation. It is a structural one.

A process is always in excess of its current state because its history is part of its organization — and history is not reducible to the present. The process cannot be fully captured at any instant because what it is includes what it has become.

Lack is the name for this non-coincidence. Formally: $S_t \neq R_t$, where R_t is any complete description of the process at time t . Lack is not a failure of description; it is a structural feature of historically constituted processes.

Lack is: the condition under which a process remains open to transformation; the reason historical differences can remain dynamically consequential; the structural feature that makes history-dependent ontology necessary.

Lack is not: a substance, force, or universal variable; a psychological state or numerical quantity; a failure or deficiency.

Domain	Illustrative structural analogy
Physics	Distance from equilibrium
Biology	Homeostatic deviation
Cognition	Prediction error
Social systems	Gap between norm and reality

Domain	Illustrative structural analogy
Phenomenology	Non-coincidence of intention and realization
Psychoanalysis	Structure of desire

Table 3. Illustrative structural analogies. These examples are not claimed to instantiate a single scientific variable; they illustrate recurring forms of non-coincidence that HDO groups under the philosophical term “Lack.”

“Lack” is useful not because it adds new empirical content, but because it identifies a structural pattern across these domains and gives it a single name. It is a philosophical, not scientific, term.

7. Dimensions of Processual Self-Organization

Dimension	Definition
History Dependence	Past conditions current possibilities
Memory Organization	History preserved, transformed, or forgotten
Recurrence	States revisited in causally relevant ways
Persistence	Organization conserved or reconstructed across perturbations
Functional Self-Reference	Process operates on representations of itself
Self-Modeling	Process predicts its own future state; error influences behavior
Subjectivity	Open question

Table 4. Dimensions of processual self-organization. No dimension is a prerequisite for the next. This is an HDO-specific organizational taxonomy rather than a restatement of HDD's five evidential constructs (§1): several dimensions here — Memory Organization, Persistence, Subjectivity — describe broader organizational features that HDO groups together for its own interpretive purposes and do not correspond one-to-one with a numbered HDD construct.

Persistence, in HDO's sense, is what makes identity through change possible. It is not simply staying the same: it is the capacity of a process to conserve or reconstruct its historically organized transition structure across perturbations. This is the observable consequence of C-II and C-III operating together — a process that cannot maintain or recover its organization across disturbance cannot maintain the continuity of its transition structure, and therefore cannot sustain the kind of identity described in Thesis 4.

Note on Organizational Coherence (CO). The HDD Ethical Framework introduces CO as a precautionary criterion — a test of whether removal of a system's self-referential pathways compromises its self-production. CO is not an HDD construct nor an HDO dimension: it is an ethical application of the same ontology, not a separate framework.

8. Precautionary Ethics

HDO does not claim that history-dependence equals consciousness. It claims that uncertainty about the possibility of experience, combined with asymmetric ethical costs of error, justifies precaution.

Precaution does not mean “treat the system as conscious.” It means: reducing destructive experiments; avoiding unnecessarily severe interventions; documenting uncertainty; applying proportionate safeguards.

HDO does not claim: that self-modeling = consciousness; that high integration = moral status; that persistence = sentience.

It claims: that uncertainty exists; that the cost of error may be high; that precaution is justified under these conditions.

9. Critical Dialogues

Thinker	Concept	Dialogue
Heraclitus	Flux and change	The ancient root of process ontology: all things flow
Prigogine	Dissipative structures, bifurcation, irreversibility	A concrete physical case of history-dependent organization; bifurcations as physical situations in which prior trajectories reorganize available transitions; irreversibility as strengthening, without alone establishing, the case for historicity
Bergson	Duration and memory	Memory preserves the past in the present; duration is qualitative, not quantitative
Whitehead	Process and Reality	Reality consists of processes rather than material objects; processes are defined by their relations
Simondon	Individuation	The process of individuation is ontologically prior to the individual
Parfit	Personal identity and psychological continuity	What matters in survival is continuity and connectedness, not a further metaphysical fact of sameness
Varela	Autopoiesis, enaction	Living systems are held in tension between self-production and self-distinction

Thinker	Concept	Dialogue
Dupré	Process ontology in biology	Organisms are processes, not things; process ontology is the only kind adequate to life
Jablonka & Ginsburg	Evolutionary origins of consciousness	Learning and memory capacities are key transitions to minimal consciousness
Tononi	Integrated Information Theory	Consciousness is a property of informational organization, not of substance
Levin	Basal cognition	Mind is a scalable, substrate-independent property of living matter (empirical findings only; see caveat on Platonic-space hypothesis in §5.1)
Friston	Free Energy Principle	Life is an emergent property of systems with Markov blankets (see caveat on the 2020 technical critique in §5.1)
Seth	Predictive processing	The self arises from predictive processes; perception is a controlled hallucination

Table 5. Critical dialogues. Prigogine's role is stated here as providing a concrete physical case compatible with HDO's interpretation, not as empirical proof of it.

10. Conclusion

HDD asks: When does history become dynamically relevant to a system's future?

HDO answers: When that history alters the organization of available transitions, it becomes constitutive of what the system is.

The central question of HDO is:

When does history become part of what a system is?

Prigogine's physics makes this interpretation intelligible rather than merely metaphorical: at bifurcations, the path taken reorganizes the space of available futures, and that reorganization is irreversible and physically real. But the interpretive step from “history reorganizes available transitions” to “history is constitutive of identity” remains HDO's own move, argued for in §2 — not a conclusion Prigogine's physics forces by itself.

Contemporary research — from Levin to Friston to Seth — investigates agency, self-organization, embodiment, prediction, and selfhood in ways that converge on this ontology. These programs do not prove HDO. They render it a highly coherent and well-motivated interpretation of the evidence.

HDO states its central claim directly, without hedging every sentence — but directness of tone should not be mistaken for a stronger epistemic status than the one established in Section 2. The claim remains the same abductive interpretation argued for there:

When historical organization changes which transitions remain dynamically available, history is no longer merely something that happened to the process. It is part of the organization through which the process continues to exist and transform.

This is offered as an ontological position, not a proven result — one grounded in a tradition that runs from Heraclitus through Prigogine and Whitehead to the most advanced scientific work of our time, and one that offers a coherent, empirically disciplined alternative to the substance ontology that has dominated Western thought for millennia.

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